CS19541 – COMPUTER NETWORKS LAB

PRACTICAL 13: DEVELOPING A PING UTILITY

AIM:

To write a Python program that checks whether a destination host responds to ICMP requests and displays the response time.

SOFTWARE REQUIRED:

Python 3, a terminal/command prompt, and network access. Raw-socket permission may require Administrator or root access.

ALGORITHM:

1. Read the destination host name or IP address.
2. Resolve the host name to an IP address.
3. Create a raw socket configured for ICMP packets.
4. Prepare an ICMP Echo Request and calculate its checksum.
5. Send the request and note the starting time.
6. Wait for a reply; calculate and display round-trip time when received.
7. If the request times out, print a timeout message; repeat and close the socket.

PROCEDURE:

1. Create a file named ping_test.py in a Python editor.
2. Type the program on page 2 and save the file.
3. Open a terminal and run the script with the required permissions.
4. Enter a host such as 8.8.8.8 and check the displayed replies or timeout messages.

WORKING PRINCIPLE:

The program sends ICMP Echo Request packets to the selected host. The time between sending a request and receiving a response is used as the round-trip

time. If no response arrives within the timeout period, a timeout message is shown.

PYTHON PROGRAM AND OUTPUT:

PROGRAM:

```
import socket, struct, time
host = input("Enter destination host or IP: ")
ip = socket.gethostbyname(host)
sock = socket.socket(socket.AF_INET, socket.SOCK_RAW,
socket.IPPROTO_ICMP)
sock.settimeout(2)
try:
for seq in range(1, 5):
# Build ICMP Echo Request header
packet = struct.pack("!BBHHH", 8, 0, 0, seq, 1)
total = 0
for pos in range(0, len(packet), 2):
total += (packet[pos] << 8) + packet[pos + 1]
total = (total >> 16) + (total & 0xffff)
checksum = (~total) & 0xffff
packet = struct.pack("!BBHHH", 8, 0, checksum, seq, 1)
start = time.time()
sock.sendto(packet, (ip, 0))
try:
sock.recvfrom(1024)
elapsed = (time.time() - start) * 1000
print(f"Reply from {ip}: {elapsed:.2f} ms")
except socket.timeout:
print(f"Request {seq}: timed out")
finally:
sock.close()
```

EXECUTION

python ping_test.py
SAMPLE TERMINAL OUTPUT (illustrative)
C:\> python ping_test.py
Enter destination host or IP: 8.8.8.8
Reply from 8.8.8.8: 24.31 ms
Reply from 8.8.8.8: 23.87 ms
Reply from 8.8.8.8: 25.02 ms
Reply from 8.8.8.8: 24.66 ms

OUTPUT:
For each response, the program prints the destination IP address and the measured time in milliseconds. If no reply arrives in time, a timeout message appears.

RESULT:
Thus, a Python-based ping utility was written to send ICMP requests and report reply times or timeout messag